

EC-350 AI and Decision Support Systems

Week 8

Introduction to Machine Learning

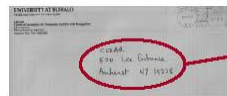
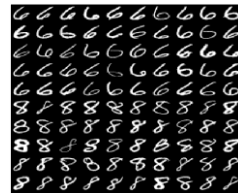
Dr. Arslan Shaukat



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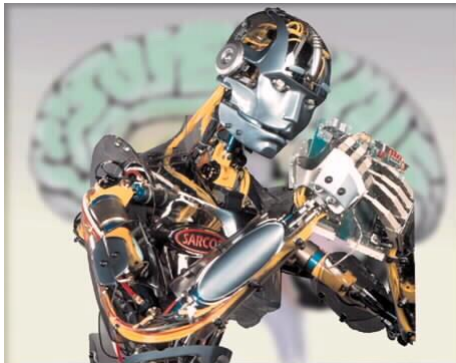
Machine Learning Applications

- Optical Character Recognition (OCR)
 - *Sorting letters by postal code*
 - *Reconstructing text from printed materials*
 - *Handwritten character recognition*
- Analysis and identification of human patterns (Biometric classification)
 - *Face recognition*
 - *Handwriting recognition*
 - *Fingerprints and DNA sequence identification*
 - *Iris scan identification*
 - *Speech recognition/speaker identification*

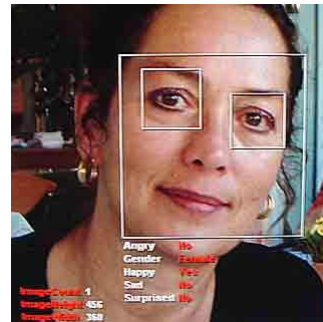


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Applications



amazon.com.



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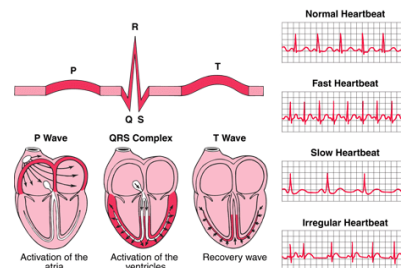
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Examples of ML

- Computer aided diagnosis
 - Medical imaging, EEG, ECG signal analysis
 - Designed to assist (not replace) physicians
- Prediction systems
 - Weather forecasting (based on satellite data)
- Information Retrieval
 - Data Mining



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Machine Perception and Pattern Recognition

- Machine Perception
 - *Build a machine that can recognize patterns*
- Pattern Recognition
 - *Theory, Algorithms, Systems to Put Patterns into Categories*
 - *Relate Perceived Pattern to Previously Perceived Patterns*
- By building such systems, we gain understanding of machine learning, particularly in humans

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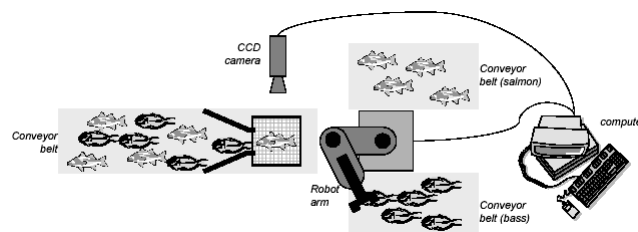
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A Machine Learning Example

- A fish processing plant wants to automate the process of sorting incoming fish according to species (salmon or sea bass)
- The automation system consists of
 - *A conveyor belt for incoming products*
 - *A vision system with an overhead camera*
 - *A computer to analyze images and control the robot arm*



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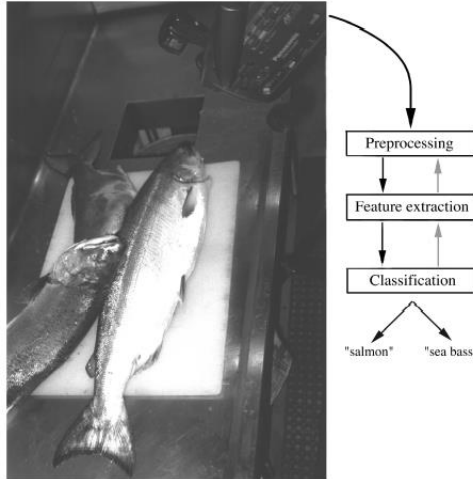
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Example

- “Sorting incoming fish on a conveyor belt according to species using optical sensing”.



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Problem Analysis

- Set up a camera and take some training sample images to extract features.
 - *Length*
 - *Lightness*
 - *Width*
 - *Number and shape of fins*
 - *Position of the mouth, etc...*
- This is the set of all suggested features to explore for use in our classifier
- Purpose:
 - *To classify the future samples based on the data of extracted features from the training samples*

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Example

- Models
 - *There are differences between sea bass and salmon and are viewed as having different models.*
- Preprocessing
 - *Segmentation*
 - *Isolate fish from one another and from the background.*
- Feature Extraction
 - *Information from a single fish is sent to a feature extractor whose purpose is to reduce the data by measuring certain features.*
- Classification
 - *Evaluates the evidence presented and makes a final decision.*

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Selection Criterion

- Suppose sea bass is generally longer than a salmon.
- Select only the length of the fish as a possible feature for discrimination.
- To choose critical value of length, we could obtain some design or training samples of the different types of fish.

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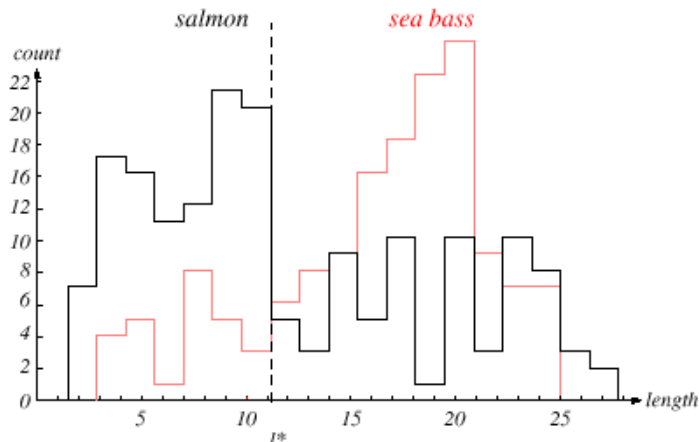
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Histograms for the Length Feature



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Selection Criterion

- No matter how we choose the threshold value of length, we cannot reliably separate sea bass from salmon.
- The length is a poor feature alone!
- Select the average lightness of the fish scales as a possible feature.

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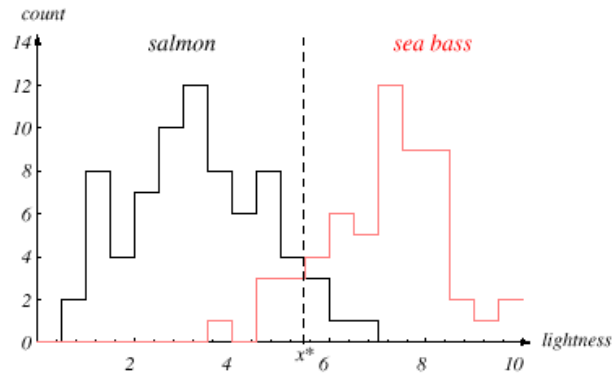
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Histograms for the Lightness Feature



- Length or Lightness, which one is a better feature?
- No value of either feature will “classify” all fish correctly